Aiden Luke Daniel

Employment/Education

2025 – Present	**University of Leeds, UK**. Research Fellow Working with Professor Jiannis Pachos, I research into the maximal thermalisation properties exhibited by 1D many-body Hamiltonians and their uses for quantum computing protocols. Remarkably, these Hamiltonians display black hole phenomena. This involves using diverse numerical techniques like Krylov subspace methods, exact diagonalisation, tensor networks, perturbation theory and Clifford circuits.
2021 – 2025	**University of Leeds, UK.** PhD in Theoretical Condensed Matter Physics, Supervisors: Prof. Jiannis Pachos, Prof. Zlatko Papic Thesis: "*Controlling, Evading, and Maximising Quantum Thermalisation*"
2017 – 2021	**University of Leeds, UK.** MPhys & BSc in Theoretical Physics , 1st Class (Honours) Supervisor: Dr. Rob Purdy Thesis: "*A Grand Clockwork SO(10) Solution*"

Research Interests

Quantum physics, quantum information, many-body physics, quantum thermalisation, quantum many-body scars, quantum computing, quantum teleportation, Gaussification, time crystals, quantum criticality, quantum gravity, SYK, matrix product states, exact diagonalisation, Krylov subspace methods, non-integrable systems, strongly correlated systems, quantum error correction, quantum machine learning, topology, optimisation, mean-field theories.

Published Papers/Preprints

- *Ergodicity breaking meets criticality in a gauge-theory quantum simulator*, Ana Hudomal, Aiden Daniel, Tiago Santiago do Espirito Santo, Milan Kornjača, Tommaso Macrì, Jad C. Halimeh, Guo-Xian Su, Antun Balaž, and Zlatko Papić, December 2025, arXiv:2512.23794

- *Quantum teleportation between simulated binary black holes*, Aiden Daniel, Tanmay Bhore, Jiannis K. Pachos, Chang Liu, and Andrew Hallam, March 2025, arXiv:2503.10761

- *Optimally scrambling chiral spin-chain with effective black hole geometry*, Aiden Daniel, Andrew Hallam, Matthew D. Horner, and Jiannis K. Pachos, March 2025, Scientific Reports, 15(1)

- *Persistent non-Gaussian correlations in out-of-equilibrium Rydberg atom arrays*, Aydin Deger, Aiden Daniel, Zlatko Papic, and Jiannis K. Pachos, December 2023, PRX Quantum, 4(4)

- *Bridging quantum criticality via many-body Scarring*, Aiden Daniel, Andrew Hallam, Jean-Yves Desaules, Ana Hudomal, Guo-Xian Su, Jad C. Halimeh, and Zlatko Papic, June 2023, Physical Review B, 107(23)

- *Hypergrid subgraphs and the origin of scarred quantum walks in the many-body Hilbert space*, Jean-Yves Desaules, Kieran Bull, Aiden Daniel, and Zlatko Papic, June 2022, Physical Review B, 105(24)

Academic Awards and Honours

- Leeds Doctoral Scholarship supported by the University of Leeds for a fully funded PhD due to Academic ability, 2021-2025

- E.C Stoner Award – Awarded for highest undergraduate classification average in the University of Leeds Theoretical physics program, 2017-2021
- Best Second Year Student – Awarded for the highest 2nd year classification average, 2018-2019
- 2nd Place in Leeds Rotary engineering competition two years in a row across 130 students across 14 schools, 2015, 2016

Talks, Conferences, and Workshops

- QIST White Rose meeting. Leeds, UK, May 2025
- APS March Meeting, Global Physics Summit 2025. Anaheim California, USA, March 2025. **TALK**: *Bridging Criticality via many-body scarring*
- QIST White Rose meeting. York, UK, October 2024
- Internal Physics Seminar Series. Leeds, UK, October 2024 **TALK**: *Quantum scarring in the detuned PXP model*
- Bridging the physics and Mathematics of Quantum many-body Chaos. Helsinki, Finland, June 2024. **TALK**: *Many-body scarring and criticality in the PXP Model*
- Internal Physics Seminar Series. Leeds, UK, April 2023 **TALK**: *Persistent non-Gaussian correlations in out-of-equilibrium Rydberg atom arrays*
- University of Leeds PGR Symposium. Leeds, UK, February 2024. **TALK**: *Thermalisation in quantum many-body physics and how to evade it*
- International Quantum Tensor Networks Meeting. Glasgow, UK, January 2024. **POSTER**: *Bridging criticality via many-body scarring*
- European Tensor Network School. Abington, UK, September 2023
- Internal Physics Seminar Series. Leeds, UK, August 2023 **TALK**: *Bridging criticality via many-body scarring*
- Quantum Transport Meeting. London, UK, September 2022. **POSTER**: *Dynamical phase diagram for many-body scarring in PXP*
- International Tensor Network Meeting. Leeds, UK, August 2022. **TALK**: *Dynamical phase diagram for many-body scarring in PXP*

Teaching

- 2021-2025: Personally supervised several BSc, MPhys and MSc students with their projects (8 in total), some of which have gone on to do PhDs
- 2024-2025: PHYS2311 – Physics 4, Quantum Phenomena
- 2023-2024: PHYS1290 – Maths 1, Scalars and Vectors. PHYS2311 – Physics 4, Quantum Phenomena. PHYS3383 – Advanced Quantum Physics (Lecturing). PHYS2050 – Communicating Physics
- 2022-2023: PHYS1290 – Maths 1, Scalars and Vectors. PHYS2311 – Physics 4, Quantum Phenomena. PHYS2050 – Communicating Physics
- 2021-2022: PHYS2311 – Physics 4, Quantum Phenomena

Mentoring and Outreach

- 2022 – 2024, A-Level maths/physics tutor, dedicated 2 hours a week to volunteer for tutoring Maths/Physics to A-level students as part of the University of Leeds Theoretical Physics group public tutoring scheme.
- 2021-2024, Ran 5 “hands-on” quantum computing workshops for visiting students/schools which introduce quantum information science and the IBM public dashboard.
- 2021 Undergraduate Mentor for the Institute of Physics, Brightside: Influenced a research trial over 3 months to improve gender balance in A-Level Physics by mentoring students. Awarded for going beyond the scope of the program and sending thoughtful messages and feedback.
- 2019 – 2020, Personal school tutor, University of Leeds, visited a local high school weekly and mentored a pupil premium student weekly with STEM.
- 2022, Work experience supervisor, University of Leeds. Supervised 4 A-level work experience students who produced and presented a poster based on my research.